3.2.6 Obstacles in chain surveying

Obstacles prevent direct measurement of distances between the points and pose problems. Hence the distances are to be found by indirect measurement.

There are three types of obstacles

1. Obstacles to ranging
2. Obstacles to chaining
3. Obstacles to both ranging and chaining

1. Obstacles to ranging

This type of obstacle in which the ends are not intervisible, but the distance between the points can be measured.

Example: rising ground, hill, wooded forest.

2. Obstacles to chaining

In this method chaining is obstructed but the two end points are intervisible. Obstacles to chaining are pond, river, lake and plantations.

There are two cases of this obstacle.

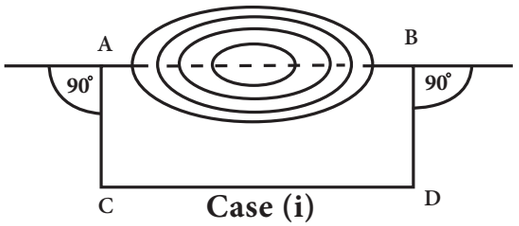
i. When it is possible to chain round the obstacle.

ii. When it is not possible to chain round the obstacle.

Case (i) When it is possible to chain round the obstacle.

In obstacles to chaining like lake, pond, hedge, etc.

1. Select two points in A & B on either side of the obstacle.
2. Erect two perpendiculars AC and BD from stations A and B. ie., AC = BD.
3. Join C and D and measure the distance between them.
4. The distance CD is equal to AB.

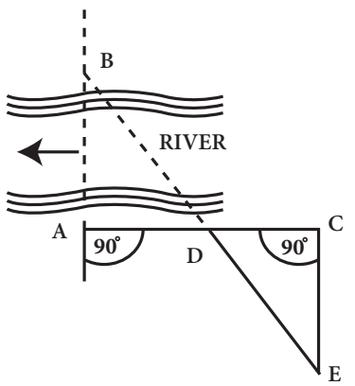


Case (i)

Case (ii) Impossible to chain round the obstacle

In obstacles to chaining like river, stream etc.

1. Select two points A and B on either side of the obstacle.
2. Erect perpendicular AC and bisect it at D.
3. Erect perpendicular CE at C and range E in line with BD.
4. Measure CE That is equal to AB (CE=AB)



Case (ii)

3. Obstacles to both chaining and ranging

In this case, the following procedure is adopted to determine the obstructed distance. A tall building is a typical example.

1. Choose A and B on one side of the obstacle.
2. Erect perpendiculars AC and BD of equal length.
3. Join CD and prolong it to pass through the obstacle.
4. Choose E and F on CD erect the EG and FH equal to that of AC and BD. Join GH and prolong it.

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